AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 30 · STUDENT LESSON

Improve Generalization

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Name the evidence. Change one plan. Keep the comparison fixed.

Test an inspectable generalization repair while separating diagnosis,development improvement and final evidence.

Keep one habit: Complete crossed-training trial → pictured background audit → four repair patterns → fixed-case comparison → independent new investigation.

MAIN LESSON · SECTION 1

Which change addresses the evidence,and what will count as help?

> Record the hypothesis,controlled factors and all outcomes before claiming improvement.

Current controlled-improvement goals

I can plan one **evidence**-led change and test it on source-separated validation cases.

I can audit crossed conditions while keeping model versions and comparison cases fixed.

I can distinguish a development improvement from a causal or final-performance guarantee.

Recall validation and source-coverage limits

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from **Overfitting**. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.
- Can a zero **training** error guarantee good validation performance?

Readiness for the shift extension

- Why does a test score need a population description?

> **My readiness explanation**

End goal

By the end, I can explain improve **generalization** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Different failure causes require different repairs: simplicity, representative data, regularization, or better features.



LEARNING TARGET All spotted **training** leaves use black backgrounds. Propose a collection and comparison that tests the background shortcut without tuning on final test examples.

Builder name and experiment date:

Readiness for the connected explanation

Readiness for Distribution shift

This week's end goal

MAIN LESSON · SECTION 2

Start with one inspectable hypothesis and one planned change

> Keep the learning procedure fixed while the declared training-data version changes.

Current generalization experiment

Improve generalization by matching a proposed change to observed evidence, not by making every component more complex at once. In this complete toy experiment, the original training labels track both a leaf marker and background colour. A fixed learner chooses one of those two available binary flags, with background first on a training-score tie.

Add crossed training conditions as one declared data-version change, then compare frozen versions on the same separate validation plants. The marker is perfectly aligned with the toy target by construction; this is not a claim about real disease features.

Distribution-shift context and scope

A model trained on one set of conditions may encounter another. Lighting, measuring tools, class frequency and the relationship between **input**s and labels can change.

A shift does not automatically imply failure, but it challenges the assumption that old **evidence** represents current use.

Use fictional classroom data throughout this investigation. Keep the task, units, **reference labels** and data split visible as you reason. Earlier weeks supplied the model-building **process**; this week examines **distribution shift**.

▼ Review · Earlier capacity explanation

A more flexible model can represent complicated relationships and accidental details. **Capacity** can help when there is enough relevant **evidence**, but perfect **training** performance is not the goal.

Compare **training** and validation using the same metric while checking leakage and representativeness.

Use fictional classroom data throughout this investigation. Keep the task, units, **reference labels** and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **overfitting** and model **capacity**.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use improve **generalization** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

Apply the connected idea: Distribution shift

MAIN LESSON · SECTION 3

Run a complete controlled training-data revision

> The saved learner and same-case comparison make the changed feature choice inspectable.

Complete original and added training cards

Training plant ID	Background red?	Available leaf marker?	Toy spotted label
T1	1	1	1
T2	1	1	1
T3	0	0	0
T4	0	0	0
T5 added crossed	0	1	1
T6 added crossed	0	1	1
T7 added crossed	1	0	0
T8 added crossed	1	0	0

Complete fixed two-plant/four-view validation

Validation view / plant group · Background red? · Marker? · Toy reference · V1 background prediction · V2 marker prediction

Validation view / plant group: P1-red /P1

Background red?: 1

Marker?: 1

Toy reference: spotted

V1 background prediction: spotted

V2 marker prediction: spotted

Validation view / plant group: P1-grey /P1

Background red?: 0

Marker?: 1

Toy reference: spotted

V1 background prediction: healthy

V2 marker prediction: spotted

Validation view / plant group: P2-red /P2

Background red?: 1

Marker?: 0

Toy reference: healthy

V1 background prediction: spotted

V2 marker prediction: healthy

Validation view / plant group: P2-grey /P2

Background red?: 0

Marker?: 0

Toy reference: healthy

V1 background prediction: healthy

V2 marker prediction: healthy

Declared candidate,tie and source-group contract

V1 fits only T1–T4; V2 fits T1–T8 after the crossed additions. Candidate flags are background_red then leaf_marker; rule predicts spotted 1 when the chosen flag is1, otherwise healthy 0. Minimize total training mistakes and keep first candidate on a tie. V1 errors[0, 0]choose background; V2 errors[4, 0]choose marker.

Training plants and validation P1/P2do not overlap, and both views of each validation plant stay together. No validation label enters fitting. Four validation views are only two new plant identities.

Complete executable toy trial

```
# Synthetic plant IDs, background_red flag, available leaf_marker flag, toy label.
original = [('T1', 1, 1, 1), ('T2', 1, 1, 1),
            ('T3', 0, 0, 0), ('T4', 0, 0, 0)]
crossed_additions = [('T5', 0, 1, 1), ('T6', 0, 1, 1),
                     ('T7', 1, 0, 0), ('T8', 1, 0, 0)]
# Two held-out plants P1/P2, each observed on both backgrounds; four views.
validation = [('P1-red', 1, 1, 1), ('P1-grey', 0, 1, 1),
              ('P2-red', 1, 0, 0), ('P2-grey', 0, 0, 0)]


def fit_feature(training):
    if not training or len({r[0] for r in training}) != len(training):
        raise ValueError('Need nonempty uniquely identified training records')
    if any(type(v) is not int or v not in (0, 1) for r in training for v in r[1:]):
        raise ValueError('Need complete binary flags and toy labels')
    # Human-declared rule: predict 1 iff the chosen flag is 1.
    # Candidate order/background-first tie policy is fixed before validation.
    candidates = [1, 2]
    errors = [sum(r[index] != r[3] for r in training) for index in candidates]
    winner = min(range(len(candidates)), key=lambda i: errors[i])
    return candidates[winner], errors


def predict(flags, feature):
    if len(flags) != 2 or any(type(v) is not int or v not in (0, 1) for v in flags):
        raise ValueError('Need both observed flags in background/marker order')
    return flags[feature - 1]


for version, training in [('V1', original), ('V2', original + crossed_additions)]:
    feature, errors = fit_feature(training)
    predictions = [predict(r[1:3], feature) for r in validation]
```

```
    correct = sum(p == r[3] for p, r in zip(predictions, validation))
    print(version, feature, errors, predictions, correct, len(validation))
# This toy marker perfectly tracks the toy label by construction, not in real use.
# Validation is visible development evidence; no untouched final test is supplied.
```

▼ Review · Earlier aggregate capacity comparison

On the same regression split, a constant has **training** MAE 5 and validation 6, a line has **training** 2 and validation 3, and a highly flexible curve has **training** 0 and validation 8. Under smallest validation MAE, choose the line.

The curve has the lowest **training** loss but the worst validation result.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

▼ Extension · Bright/dim shift result and limits

On development-like bright images, a frozen classroom classifier makes 2 errors in 20 cases:10 percent. On a separate dim set it makes 6 errors in 20:30 percent. The pooled result is 8/40=20 percent error.

The pooled score hides a 20-percentage-point difference between conditions.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

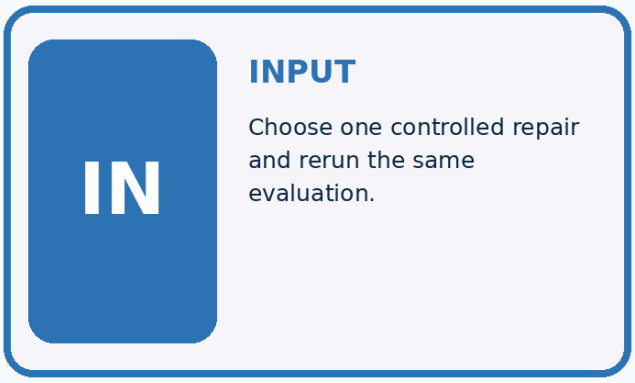
Keep the distinction between an observed result and a proposed next step.

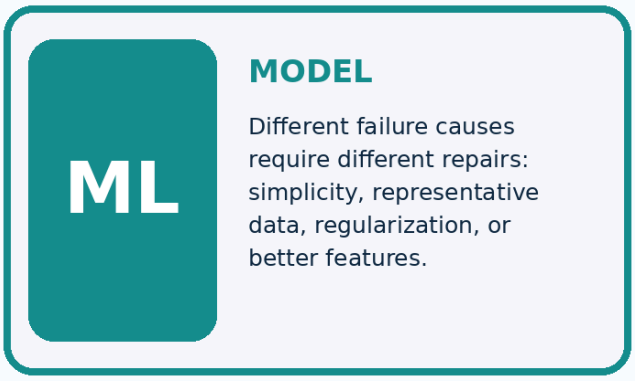
▼ Earlier wording, alternatives and picture transcripts

ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Choose one controlled repair and rerun the same evaluation.

MODEL IDEA Different failure causes require different repairs: simplicity, representative data, regularization, or better features.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD **Evidence**: high train score, low validation score, largest gap in low light. Hypothesis combines complexity and missing low-light coverage; choose one cause to test first.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Apply the connected idea: Distribution shift

Connect the picture and the explanation

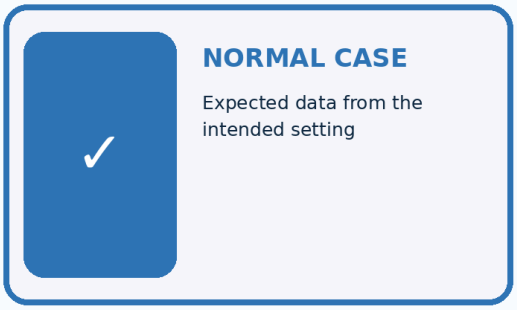
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

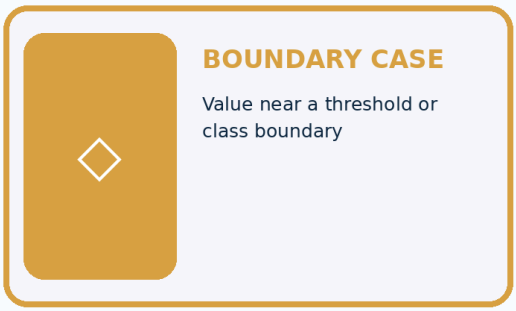
Compare a targeted repair with a stronger claim than its evidence supports

> A revised version needs both old and new case-level results.

Four evaluation conditions



NORMAL Expected data and expected behavior.

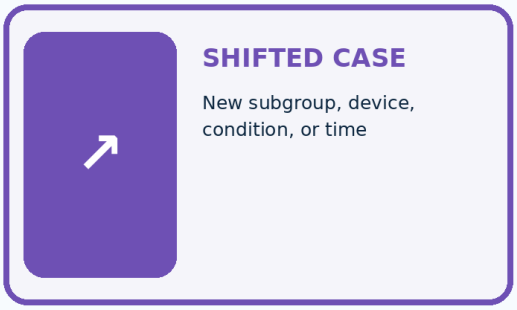


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves **generalization**. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes.

▼ **Check after predicting**

Prediction: It fails on spotted/grey and healthy/red in the crossed audit.

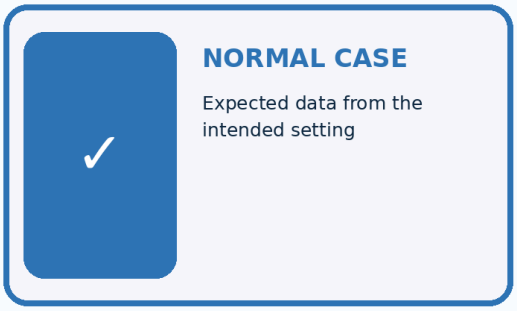
Reason: Those supplied **prediction**s follow background rather than the intended leaf target.

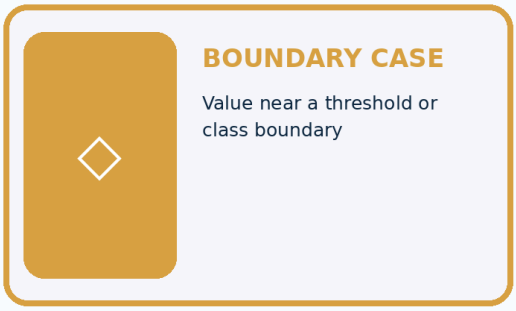
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

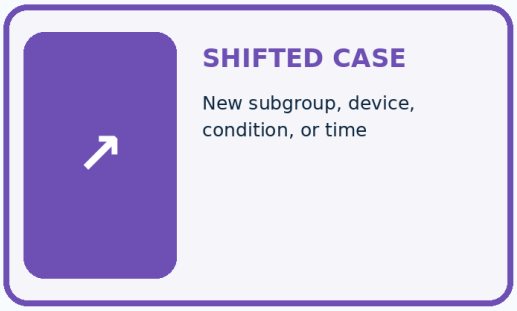
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Cross the class and background before naming the shortcut

> Keep the target leaf and model version fixed while auditing background sensitivity.

Complete pictured facts and trial scope

Test whether a plant model follows the leaf or a convenient background shortcut.

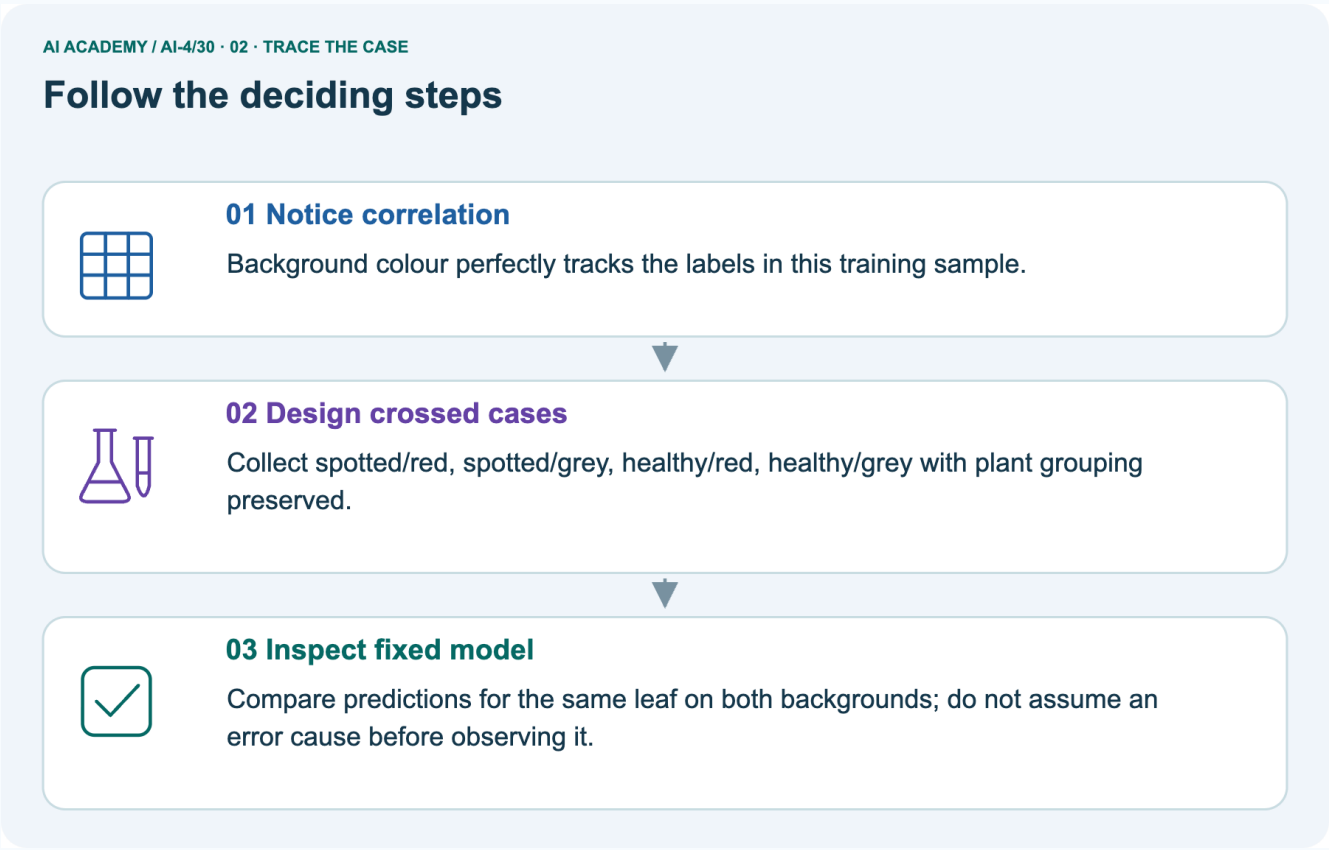
Toy **training** photos show spotted leaves on red cards and healthy leaves on grey cards. Intended use allows both backgrounds for either class.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The pictured red/grey audit and current executable trial share the intended four class/background combinations. Under the supplied background-following prediction pattern, spotted/grey and healthy/red fail while spotted/red and healthy/grey match:2/4 views, but only 2 plant identities in the complete trial.

Adding crossed training cards changes this particular learner's selected feature and yields 4/4 development views. The perfect toy marker is constructed, not a real diagnostic feature; these reused development views and small source count do not establish final performance or how every internal model representation works.

Trace crossed conditions and fixed-model audit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Notice correlation	Background colour perfectly tracks the labels in this training sample.
2. Design crossed cases	Collect spotted/red, spotted/grey, healthy/red, healthy/grey with plant grouping preserved.
3. Inspect fixed model	Compare predictions for the same leaf on both backgrounds; do not assume an error cause before observing it.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

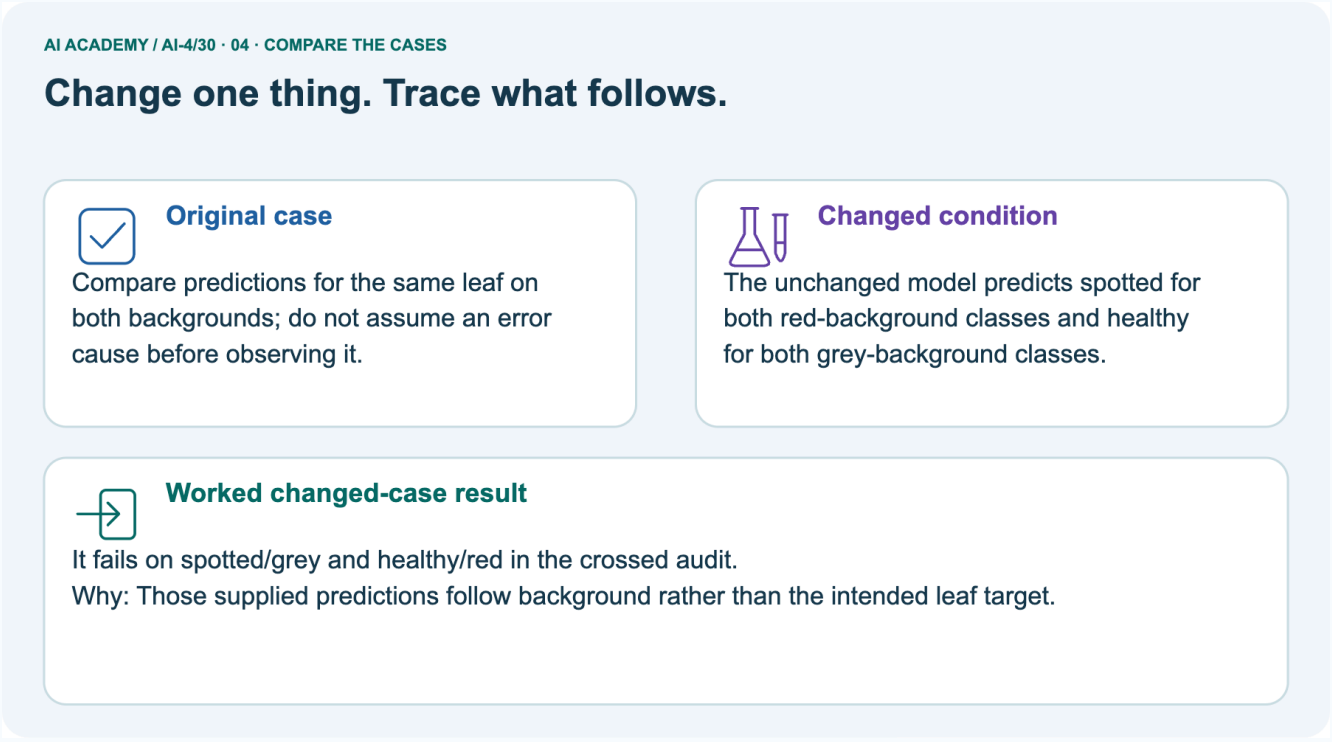
Read the picture in order. Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the supplied background-following outputs

- Change one thing: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain fresh-evidence and representation limits

- One controlled audit can reveal a shortcut in this toy setting without proving how every internal feature is represented.

- Read the labelled picture: What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Extension · Separate bright/dim error rates

Distribution shift at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Bright condition	2 errors of 20	10 percent error
Dim condition	6 errors of 20	30 percent error
Combined	8 errors of 40	20 percent error

The comparison identifies an observed weakness, not its cause. Check how images and labels were collected and whether other factors changed with lighting. An input shift can be detected before labels arrive, but performance cannot be measured from predictions alone.

Why is 8/40 incomplete by itself? Explain the steps or evidence behind your response.

▼ Review · Earlier capacity gap question

Overfitting and model capacity at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Constant	Training 5	Validation 6
Line	Training 2	Validation 3
Flexible curve	Training 0	Validation 8

The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

Does a training-validation gap alone prove overfitting? Explain the steps or evidence behind your response.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 30 • WORKED EVIDENCE

1. Evidence: high train score, low validation score, largest gap in low light.
2. Hypothesis combines complexity and missing low-light coverage; choose one cause to test first.
3. Add representative low-light training examples while freezing model settings.
4. Rerun all validation cases; compare overall, error slice, subgroup gap, and regressions.

STEP 1 **Evidence**: high train score, low validation score, largest gap in low light.

STEP 2 Hypothesis combines complexity and missing low-light coverage; choose one cause to test first.

STEP 3 Add representative low-light **training** examples while freezing model settings.

STEP 4 Rerun all validation cases; compare overall, error slice, subgroup gap, and regressions.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

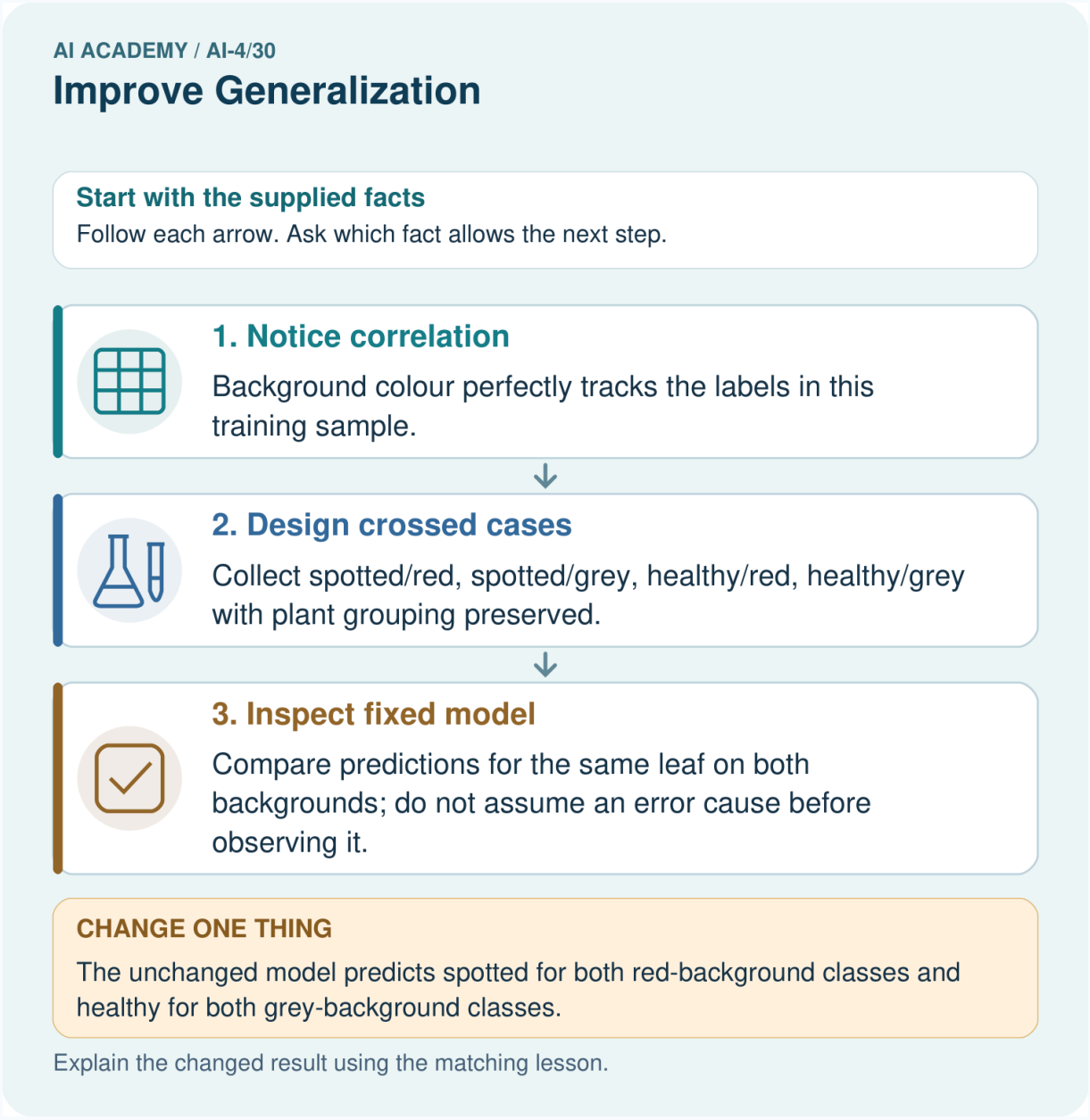
Apply the connected idea: Distribution shift

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Improve Generalization

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Separate observed correlation,diagnostic outputs and a final claim

> State exactly what the controlled record establishes.

Two complete claims to repair

Repair claim: all spotted training photos used red backgrounds, so the model must rely only on red. The correlation alone does not establish that. In the supplied crossed audit, the same leaf changes prediction with background under the frozen V1 rule, which is specific diagnostic evidence. Repair claim: V2 gets4/4 views, so it is reliable on all plants.

There are only 2 validation plants and this is inspected development evidence; new intended-use groups and reference-quality checks remain necessary.

▼ Extension · Versioning and unsupported-input response

Do not normalize away a shift by silently changing pre**process**ing during evaluation. Treat a revised transformation or retrained model as a new version. Use fresh appropriate **evidence** for its claims and retain earlier results.

> **State a safe classroom response to inputs outside the supported range.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Changing data, features, model, and threshold together is the fastest proof of improvement. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE Changing data, features, model, and threshold together is the fastest proof of improvement.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

A simpler model is not always better. If a flexible candidate has better valid **evidence** under the declared rule, keep that **evidence** visible.

Model **capacity** is one part of an experiment, alongside features, sample size, label quality and the split.

What extra evidence would you inspect before concluding a model memorized noise?

Apply the connected idea: Distribution shift

MAIN LESSON · SECTION 8

Practise explaining the controlled change with less help

> Report both the gain and the remaining evidence gaps.

Current task and response

- For the complete T1–T8 and P1/P2 fixture, explain V1 and V2 candidate training errors, chosen feature and every validation prediction. Report both view and plant counts. State what was held fixed, what changed and why 4/4 development views do not prove general reliability.

My controlled-improvement reasoning and evidence

State pictured fresh-evidence needs

- What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Keep independent work separate

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Extension · Declared shift fallback

For a supervised classroom prototype, define a review trigger before use, such as a new unsupported **input** range or a documented increase in labelled error.

Pause the affected use, provide a human or non-model **fallback**, and investigate with new development **evidence**. Version any revision.

▼ Review · Earlier capacity practice

Change one development factor at a time: simplify the candidate, improve data coverage or adjust a declared regularization setting. Refit on **training** and compare on validation.

Do not remove difficult valid examples or repeatedly search the test for a favorable result.

Use workbook W2 for the connected example "**Overfitting** and model **capacity**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

▼ Extension · Earlier shift practice

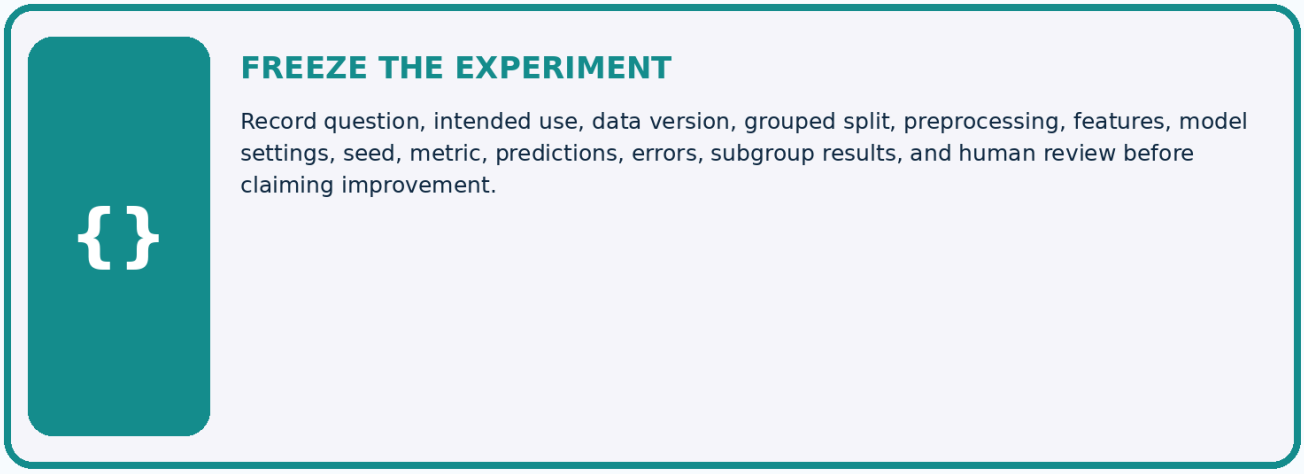
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A final answer is enough

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Apply the connected idea: Distribution shift

Practise with less help: Interpret a gap

Check your reasoning

Practise the connected topic: Distribution shift

Explain the picture

MAIN LESSON · SECTION 9

Run one complete revision and plan three other targeted repairs

> The lesson supplies the data and executable learner for the promised one-change trial.

Four complete evidence patterns and targeted plans

Complete Repair Planner evidence pattern · One targeted next change/test, with its limit

> **Complete Repair Planner evidence pattern:** 1 Original T1–T4 class/background correlation; frozen V1 changes same-leaf outputs across backgrounds
>
> **One targeted next change/test, with its limit:** Add the supplied crossed T5–T8 training conditions with learner/settings fixed, then compare on unchanged development P1/P2. Only this toy mechanism and inspected cases are demonstrated.

> **Complete Repair Planner evidence pattern:** 2 Supplied model training 100%, validation 60%, with conditions and labels not fully documented

One targeted next change/test, with its limit: Audit source/label/split differences before attributing the gap to capacity; then compare one simpler setting under a fixed protocol. No cause is proved by the two percentages.

Complete Repair Planner evidence pattern: 3 Declared label rule says touching counts positive, but 5of10 checked development labels treated touching as negative

One targeted next change/test, with its limit: Review evidence and annotation policy, version justified corrections, refit on corrected training and compare every candidate on the same reviewed validation references. Do not choose labels to match a favored model.

Complete Repair Planner evidence pattern: 4 A candidate feature is the reviewer's final decision recorded after prediction time

One targeted next change/test, with its limit: Remove the unavailable answer-derived input and refit under a valid prediction-time contract; retain the earlier leaking result as invalid evidence, not a baseline improvement target.

Normal,tied-fit,missing-input and changed-cohort probes

Complete trial probe	Expected bounded check
Normal original/crossed training versions	V1 choosesbackground; V2 choosesmarker; validation 2/4→4/4 views, 2 plants.
Boundary training-candidate tie on original data	Both zero mistakes; predeclared background-first policy decides, not validation labels.
Missing marker on proposed new query	Violates this binary observed-input contract; request/review, not invented 0.
Changed validation plants or background at the same time as the model revision	Record changed cohort/protocol; do not attribute the entire difference to the intended training addition.

Match,run or trace,and bound the conclusion

1. Repair Planner: match all four supplied evidence patterns to one justified next test and say what competing explanation remains. Execute or trace the complete T/Vtrial as the one implemented training-data change:predict candidate errors and all outputs first, preserve every old/new result, and report two plants rather than four independent plants. The literal label 1 means toy spotted; do not infer disease diagnosis.

   No new final-evaluation cases or results are supplied. Keep the selected blue/white background investigation for an uncoached first design attempt.

Keep the evidence ledger

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Missing
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

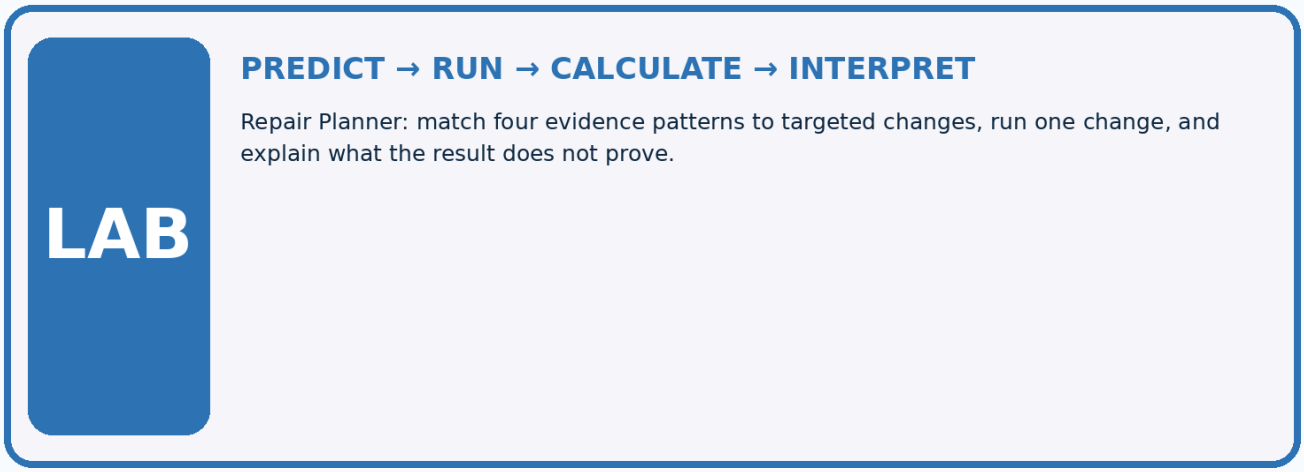
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Repair Planner: match four **evidence** patterns to targeted changes, run one change, and explain what the result does not prove.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: design a fresh background-generalization investigation

> Keep diagnosis,development repair and final evidence as separate stages.

Complete independent scenario and scope

Spotted leaves have blue backgrounds and healthy leaves white backgrounds in **training**. Design a generalisation investigation with fresh evaluation.

This selected design task uses blue/white training backgrounds, not the completed red/grey toy trial. The training correlation alone does not prove reliance on background. Plan same-leaf crossed-background diagnostics on development sources, one justified training revision, and a separate fresh grouped final evaluation after choices are frozen.

No observed blue/white predictions, benefit or final metrics are supplied; record them as proposed rather than invent results.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Changed independent case

Week 30 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A team proposes more representative **training** examples and a simpler model. How should it compare changes while protecting the final test?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result
</details>

Write or draw your reasoning and evidence.

Connect to your project

Improve Plant Classifier with one approved change and a full before/after **evidence** table.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

All spotted **training** leaves use black backgrounds. Propose a collection and comparison that tests the background shortcut without tuning on final test examples.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Keep the project repair linked to a specific uncertainty

> A report should name the controlled factors as well as the improved score.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Improve Plant Classifier with one approved change and a full before/after **evidence** table.

Save the exact hypothesis, data additions or correction, unchanged settings, source-group membership and full before/after outputs. Preserve all gains and regressions; keep missing/unsupported inputs and fallback policy visible. A targeted change is a development experiment, not approval for general use.

Freeze the revised system before a later fresh final comparison and retain the history of which cases informed development.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with one changed factor and the evidence still missing

> Explain why a convincing toy repair remains a bounded development result.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Compare **training** and validation **evidence** for model **capacity**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
generalization	performance beyond training examples	
controlled change	one planned modification	
ablation	remove one element to measure its effect	
regression test	protect previously passing behavior	

Closer-explanation terms

Word	Meaning
Capacity	Flexibility of a model to represent patterns
Overfitting	Learning details that fail to generalize well
Underfitting	Failing to capture useful structure

Word	Meaning
Generalization	Performance on relevant unseen examples

▼ Extension vocabulary

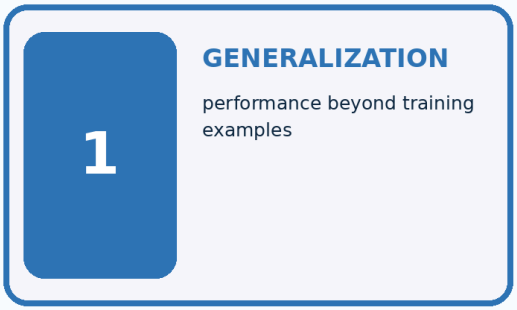
Term	Meaning
Distribution shift	A change in the data-generating conditions
Monitoring	Repeated checks of inputs, outputs and later outcomes
Reference labels	The labels used to evaluate predictions
Fallback	A planned alternative when the model is unsuitable

▼ Earlier wording, alternatives and picture transcripts

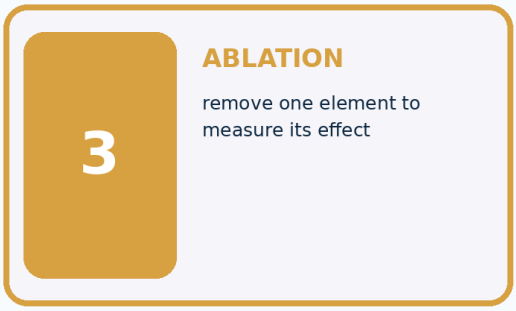
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Changing data, features, model, and threshold together is the fastest proof of improvement.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Terms for Distribution shift

▼ Extra challenge · Build a stronger explanation

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **generalization**, **controlled change**, **ablation**, **regression test**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Changing data, features, model, and threshold together is the fastest proof of improvement.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

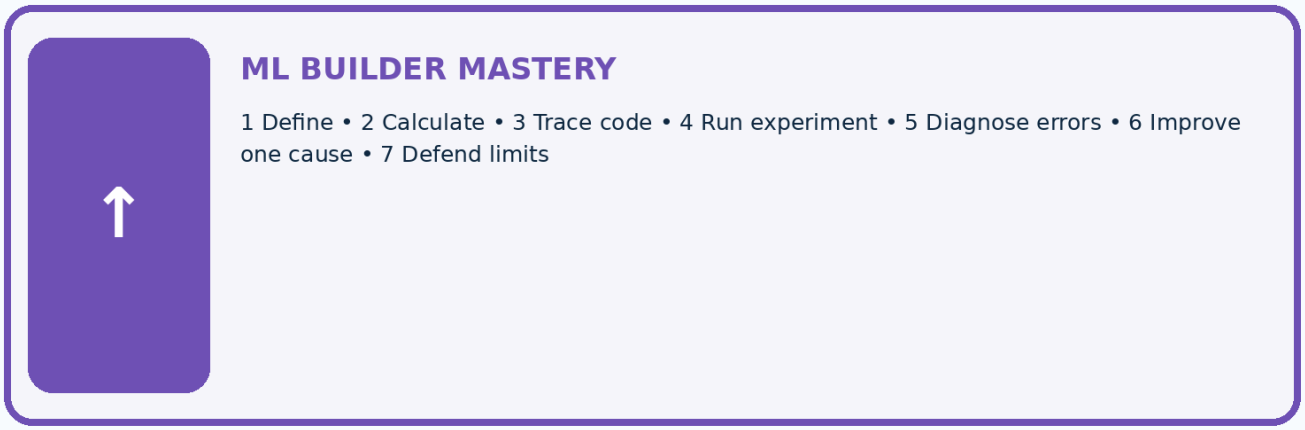
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected new blue/white background investigation design for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

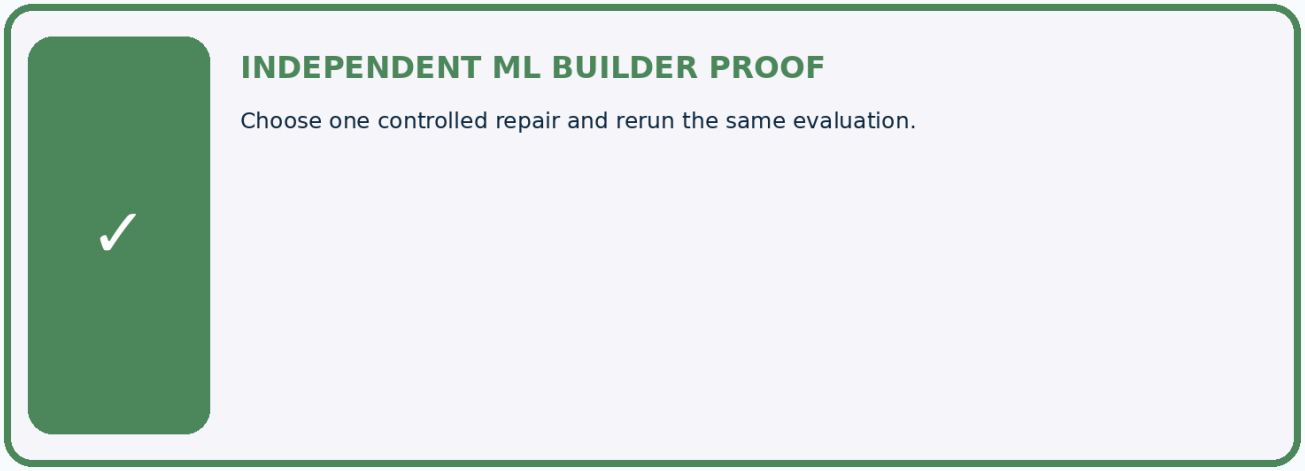
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Choose one controlled repair and rerun the same evaluation.

NEXT BRIDGE Carry the experiment **evidence** into: Subgroup Performance.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

A classifier relies on tray colour; new trays reduce performance.

Propose controlled improvement.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

<details>
<summary>Check after trying</summary>

Reduce the shortcut and add representative variation; compare on fixed validation cases including new colours. Change one planned factor.

Evidence or stage	Value or result
Observed shortcut	Tray colour tracks label
Changed condition	New trays reduce performance
Revision	Reduce shortcut and add relevant variation
Comparison	Same validation conditions and slices

</details>

▼ Earlier wording, alternatives and picture transcripts

Week 30 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Repair a background shortcut deliberately

A classifier relies on tray colour; new trays reduce performance.

Propose controlled improvement.

Reduce the shortcut and add representative variation; compare on fixed validation cases including new colours. Change one planned factor.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Compare training and validation evidence for model capacity	Explain the deciding step, show a trace or test, and state one limit.
Explain overfitting without assuming every gap proves it	Explain the deciding step, show a trace or test, and state one limit.
Use a declared selection rule instead of perfect training fit	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Test whether a plant model follows the leaf or a convenient background shortcut.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Test whether a plant model follows the leaf or a convenient background shortcut.

- Which labelled step matches this detail? Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.
- 1. Notice correlation
- 2. Design crossed cases
- 3. Inspect fixed model
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Inspect fixed model Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. Read the picture in order.

Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes. Which conclusion is supported by the supplied facts?
- 1. It fails on spotted/grey and healthy/red in the crossed audit.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: It fails on spotted/grey and healthy/red in the crossed audit. Those supplied **prediction**s follow background rather than the intended leaf target.

Repair and keep the record

Review Feature Engineering (ai-4/29). Goal: Create features available at **prediction** time. Short practice: What should be saved so another group transforms the query identically?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Overfitting** (ai-4/27). Goal: Compare **training** and validation **evidence** for model **capacity**. Short practice: Does100% **training** accuracy prove Large was always best?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B's zero **training** MAE not settle the original choice?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

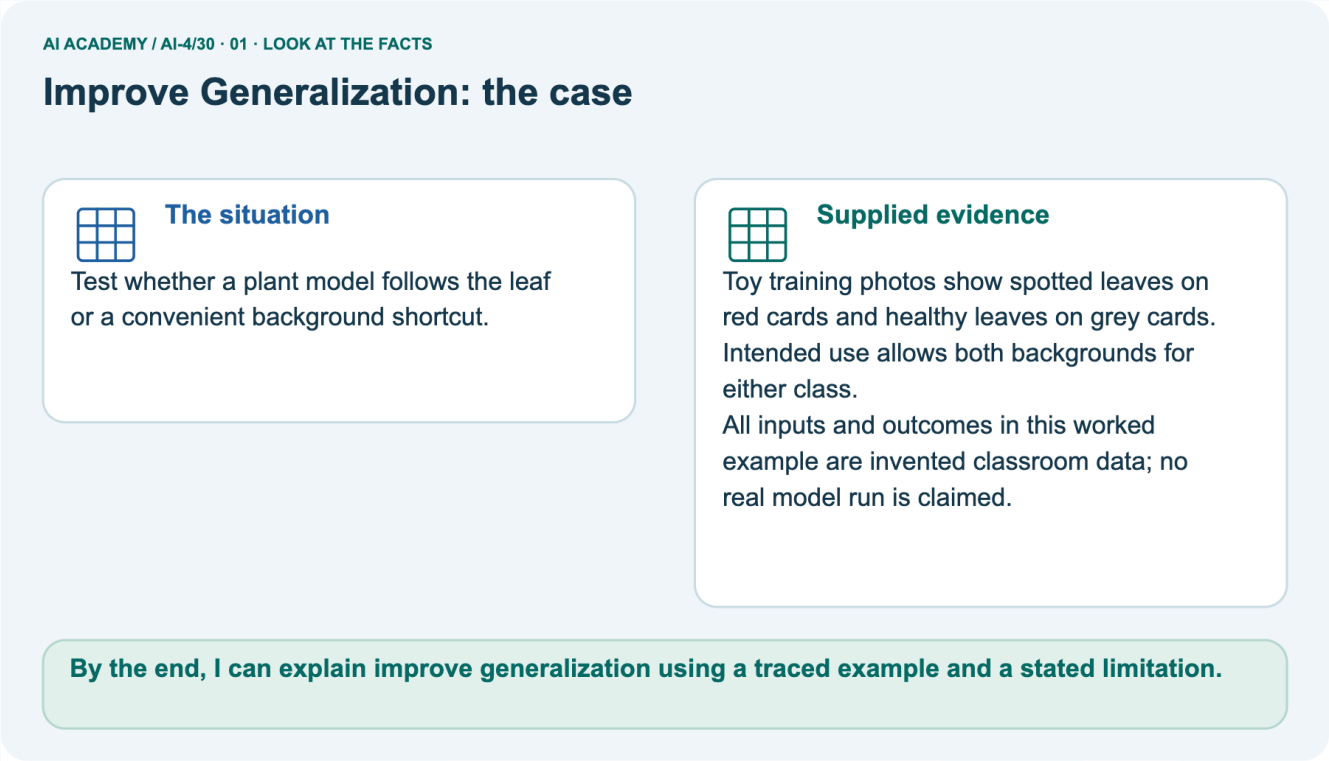
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Improve Generalization: the case



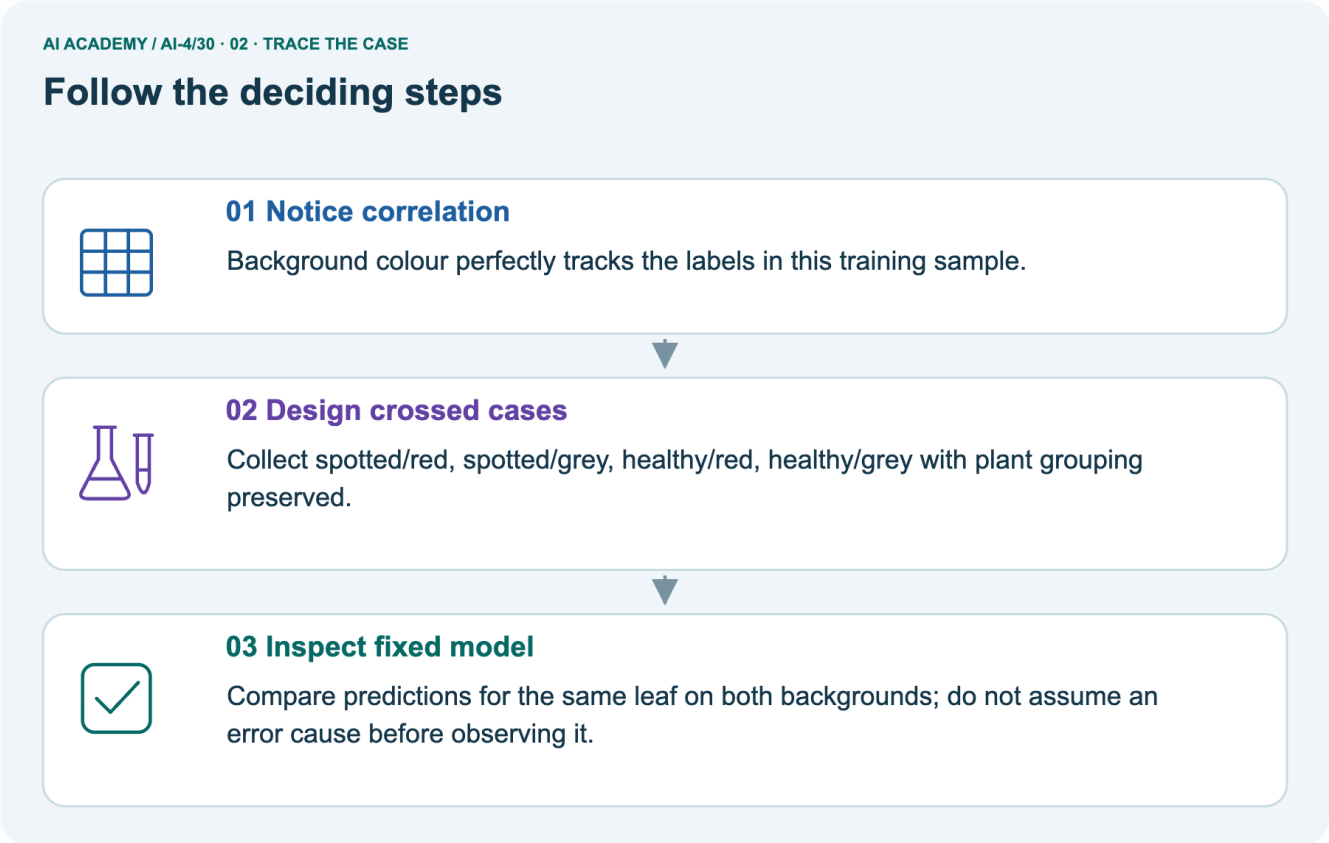
Follow the labelled picture with the matching explanation. Open the vector diagram

Test whether a plant model follows the leaf or a convenient background shortcut. Supplied facts: Toy **training** photos show spotted leaves on red cards and healthy leaves on grey cards.

Intended use allows both backgrounds for either class. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

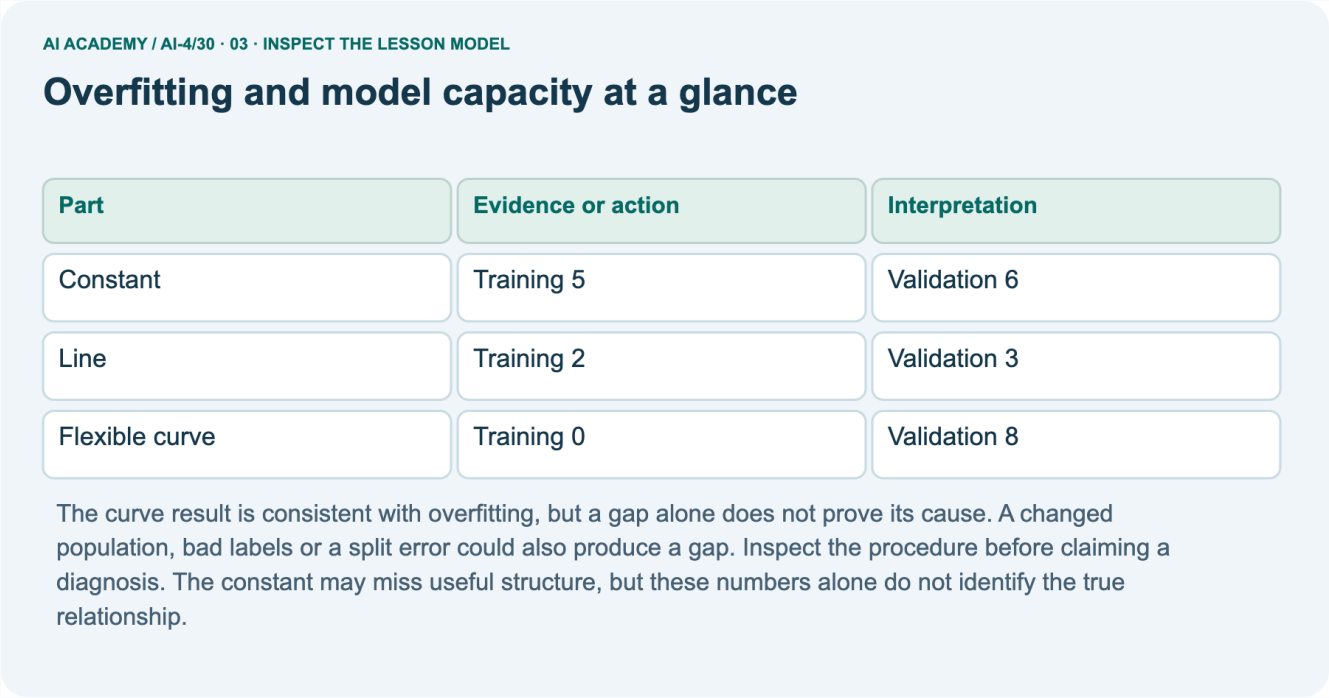


Follow the labelled picture with the matching explanation. Open the vector diagram

Notice correlation: Background colour perfectly tracks the labels in this **training** sample. → Design crossed cases: Collect spotted/red, spotted/grey, healthy/red, healthy/grey with plant grouping preserved. → Inspect fixed model: Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.

Point to each numbered stage and explain which recorded **input** it uses.

03. Overfitting and model capacity at a glance

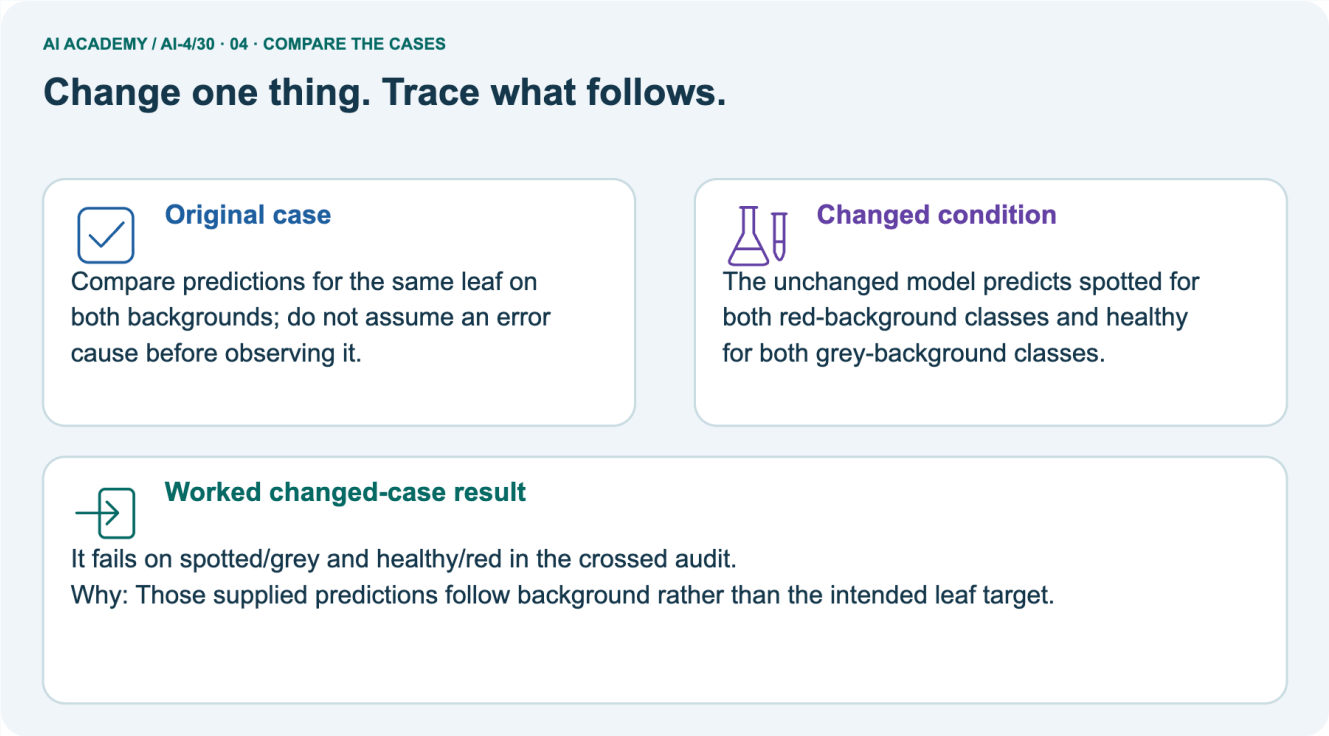


Follow the labelled picture with the matching explanation. Open the vector diagram

Overfitting and model **capacity** at a glance. Constant | **Training** 5 | Validation 6. Line | **Training** 2 | Validation 3. Flexible curve | **Training** 0 | Validation 8

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

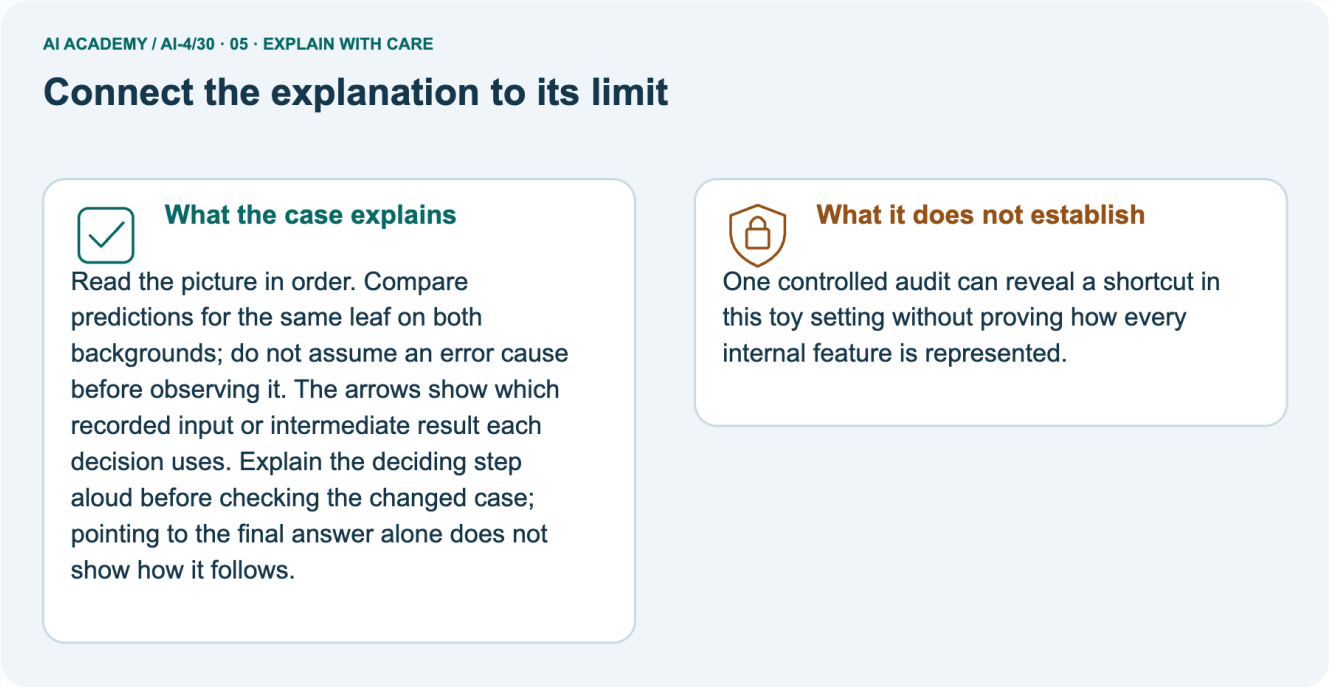
Original case: Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.

Changed condition: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes. Expected result: It fails on spotted/grey and healthy/red in the crossed audit.

Reason: Those supplied **prediction**s follow background rather than the intended leaf target.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

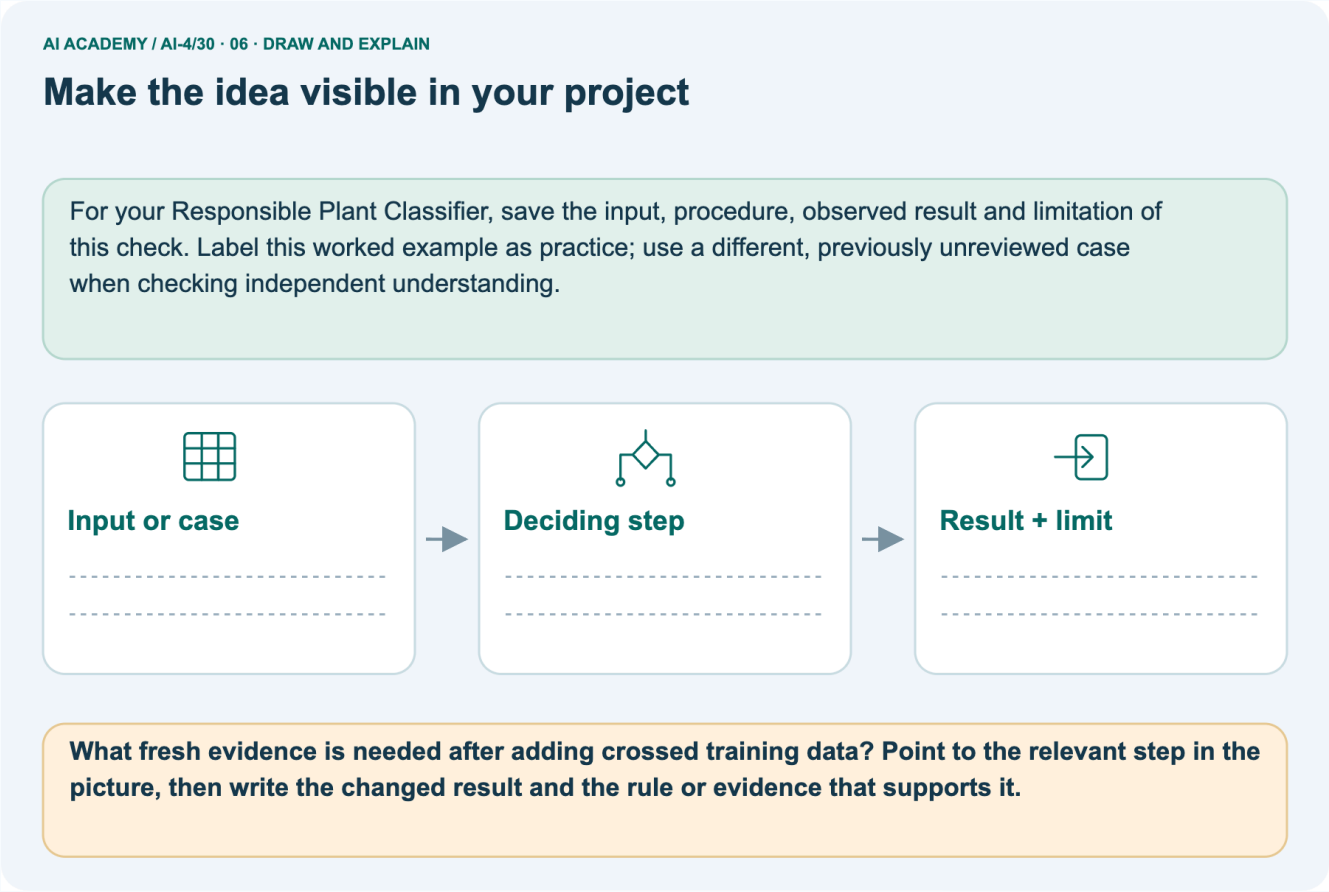
Read the picture in order. Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: One controlled audit can reveal a shortcut in this toy setting without proving how every internal feature is represented.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Improve Generalization: the case

02. Follow the deciding steps

03. Overfitting and model capacity at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.